SELECTION, NOT SCALE: WHAT CONTROLS ACCURACY AND FAITHFULNESS IN INHERENTLY INTERPRETABLE TIME SERIES CLASSIFICATION

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ABSTRACT

Inherently interpretable time series classifiers built on Multiple Instance Learning (MIL) return a label and, from the same computation, an importance score for every time step. Progress in this line is usually reported as a stronger backbone paired with a better faithfulness score. We ask a different question: inside this family, what actually controls accuracy and explanation quality? We train 72 encoder $\times$ pooling-head combinations under one identical configuration, on a dataset with per-time-step ground truth and on 85 UCR datasets, and report four findings. **(i)** Explanation quality is governed by *selection inside the aggregator* rather than by encoder capacity: restricting the bag score to the largest κ fraction of per-step evidence – a head that reduces *exactly* to the standard conjunctive baseline at $\kappa = 1$ – raises ground-truth ranking quality from 0.722 to 0.777 NDCG@n, and costs about one point of accuracy. **(ii)** Faithful interpretability is cheap: a 41 K-parameter model matches or exceeds a 424 K-parameter MILLET baseline on accuracy, AOPCR and NDCG@n *simultaneously*, and the residual gap on UCR is a capacity effect, not an architectural one – at comparable capacity our encoder outranks the baseline (11.53 against 12.60 mean rank). **(iii)** AOPCR, the faithfulness metric this literature reports, is not comparable across training configurations: the same architecture and the same code score 2.57 under one recipe and 13.27 under another, a $5.2\times$ swing that no change in explanation quality accounts for. We show why, and give a reporting protocol. **(iv)** Pooling heads that *provably* contain the baseline as a special case still fail badly in practice – the worst head we test is one of them – because nothing in the objective drives the model back to the safe branch. Capacity-safety is not optimisation-safety. Together these results suggest the aggregator, not the backbone, is where the design effort in interpretable time series classification belongs.

1 INTRODUCTION

A time series classifier that only returns a label is of limited use where someone has to act on its output. When a model flags an anomaly in a patient recording or in a machine's vibration trace, the operator needs to know *which part of the signal* triggered the alert. Rudin (2019) argues that in such settings a model that is interpretable *by design* should be preferred to a black box paired with a post-hoc explainer, because only the first guarantees that the explanation and the decision are the same computation.

For time series, the leading realisation of that idea is Multiple Instance Learning (MIL). MILLET (Early et al., 2024) treats a series as a bag of time steps, replaces the global average pooling that ends every standard convolutional classifier (Wang et al., 2017; Ismail Fawaz et al., 2020) with a MIL *pooling head* that scores each step, and computes the class prediction from those scores. The scores are therefore the explanation and the prediction at once. Work in this line is normally presented the same way: a backbone, a pooling head, an accuracy table, and a faithfulness score.

This paper asks a question that presentation leaves unanswered: *within this family, what actually controls accuracy and explanation quality?* The question is answerable because the family factorises cleanly. A model is an encoder followed by a pooling head, the two communicate through one fixed tensor interface, and either can be replaced without touching the other. We exploit that to run a controlled sweep – 72 encoder $\times$ pooling-head combinations, up to 129 datasets each, every model trained under a single identical configuration, with the MILLET baseline re-trained inside the same harness rather than quoted from its paper. Under that design, any difference in the results is attributable to the component we changed.

The sweep is built around SEA-Net, a compact multi-scale depthwise-separable encoder family and seven MIL pooling heads, each head targeting a specific limitation of MILLET's conjunctive pooling (Section 3). SEA-Net is the *instrument* of this study rather than its result: it exists so that capacity, aggregation rule and training budget can be varied independently. Four findings follow, and they are the contribution of this paper.

- **Selection inside the aggregator, not encoder capacity, drives explanation quality** (Section 5). Averaging only the largest κ fraction of per-step evidence makes the importance map exactly zero wherever the model used nothing. Because $\kappa = 1$ recovers the conjunctive baseline *exactly*, sweeping κ is a controlled experiment on selection alone: ground-truth ranking quality rises from 0.722 to 0.777 NDCG@n, with an optimum rather than a monotone gain, and costs about one accuracy point. A second, independent knob – an attention-entropy penalty – moves both quantities the same way.
- **Faithful interpretability does not require a large backbone, and the residual gap is capacity** (Section 6). A 41 K-parameter model beats the 424 K-parameter baseline on accuracy, AOPCR and NDCG@n at the same time on the one dataset where explanation correctness can be checked against ground truth. On the 85 UCR datasets the small models fall behind the baseline, but a wider version of the *same* encoder overtakes it (11.53 against 12.60 mean rank), which locates the effect in capacity rather than in architecture.
- **AOPCR is not comparable across training configurations** (Section 7). Holding the architecture, the code, the metric implementation and the data fixed and changing only the training recipe moves AOPCR from 2.57 to 13.27. AOPCR is an area over a curve of raw confidence differences, so its scale follows the scale of the logits. Every cross-paper AOPCR comparison in this literature is therefore uninterpretable, including comparisons against the published numbers of the method we build on. We give a reporting protocol that costs nothing to adopt.
- **Capacity-safety is not optimisation-safety** (Section 8). Five of our seven heads provably reduce to the conjunctive baseline for one setting of their parameters, an argument used routinely to claim a design "cannot hurt". One of them is nonetheless the worst head we test, by a wide margin. The mechanism is identifiable: no term in the objective pushes the blend parameter back towards the safe branch, so a head that *can* represent the baseline never finds it.

What we do not claim. We do not claim state of the art on the UCR archive; we are not, and Section 6 reports the margin by which. We do not claim that the encoder variants we build are individually important – Section 6 shows the best pair is not composed of the two individually best halves. And because of Section 7, we do not compare any AOPCR value in this paper against an AOPCR printed in another one. Every number we report was produced under a single configuration inside one harness, and the code that produced it is available at `https://anonymous.4open.science/r/seanet-ANON`.

2 Background

2.1 Time series as bags of time steps

A univariate series $\mathbf{X} = (x_1, \ldots, x_T)$ carries one label $y \in \{1, \ldots, C\}$, but in most datasets only a few time steps hold the evidence for it. That is exactly the structure of Multiple Instance Learning (Dietterich et al., 1997), where a labelled *bag* contains unlabelled *instances*: take the series as the bag and each time step as an instance, and recovering the explanation becomes the standard MIL problem of recovering instance-level contributions from a bag-level label.

An encoder maps every step to an embedding, $\mathbf{z}_j = f_\theta(\mathbf{X})_j \in \mathbb{R}^d$, and a MIL pooling head reduces the T embeddings to a bag prediction $\hat{y} \in \mathbb{R}^C$. Attention-based MIL (Ilse et al., 2018) learns how much each instance counts, and because the attention weight is the quantity the prediction is built from, it can be read directly as an importance score. The most active application of this idea is histopathology, where a slide is a bag and a patch is an instance; two methods from that literature – multi-scale fusion (Hashimoto et al., 2020) and the dual-stream critical-instance mechanism of Li et al. (2021) – are adapted to time series in Section 3.

2.2 The baseline: conjunctive pooling

MILLET (Early et al., 2024) pairs an InceptionTime encoder (Ismail Fawaz et al., 2020) with *conjunctive pooling*, which combines one attention gate with a per-time-step classifier:

$$a_j = \sigma\big(\mathbf{w}_a^\top \tanh(W_a \mathbf{z}_j)\big) \in [0,1], \qquad \mathbf{y}_j = W_c \mathbf{z}_j + \mathbf{b}_c \in \mathbb{R}^C, \tag{1}$$

$$\hat{y}^{\,k} = \frac{1}{T}\sum_{j=1}^{T} a_j \mathbf{y}_j^{\,k}, \qquad \mathbf{I}_{k,j} = a_j \mathbf{y}_j^{\,k}. \tag{2}$$

The interpretation $\mathbf{I}_{k,j}$ is made of the same terms that are summed to give $\hat{y}^{\,k}$, so the explanation cannot disagree with the prediction: it *is* the prediction, decomposed. Three properties of Equation (2) bound what the head can express, and each one motivates a family of heads in Section 3. **L1, one gate for all classes:** a_j is a single scalar per step, so a step cannot support one class while arguing against another. **L2, weights not normalised over time:** the gate is a sigmoid and the sum is divided by T, so the magnitude of $\hat{y}^{\,k}$ depends on the series length – awkward on an archive whose lengths span 15 to 2,844. **L3, a fixed mean:** evidence concentrated in a few steps of a long series contributes a vanishing share of the average, and a fixed operator cannot adapt to datasets where the evidence is instead spread out.

2.3 Measuring explanation quality

We use the two metrics of Early et al. (2024), computed with their implementation, because our purpose is to compare against their method on their own terms.

AOPCR (faithfulness to the model). Steps are ranked by the model's own interpretation, masked one at a time in that order, and the confidence in the original prediction is tracked; a faithful ranking makes it fall fast. Writing O^k for the induced ranking and $\mathbf{X}_j^{O^k}$ for the series with its j most important steps masked (Samek et al., 2017),

$$\mathrm{AOPC}(\mathbf{X}, O^k) = \frac{1}{T-1}\sum_{j=1}^{T-1}\Big[\hat{y}^k(\mathbf{X}) - \hat{y}^k\big(\mathbf{X}_j^{O^k}\big)\Big], \tag{3}$$

and AOPCR reports the margin of Equation (3) over three random maskings, so that a merely fragile model is not rewarded. AOPCR needs no ground truth and is therefore available on every dataset. It is also, as Section 7 shows, **unnormalised** – it is built from raw confidence differences, so its scale follows the scale of the logits.

NDCG@n (correctness against ground truth). Normalised Discounted Cumulative Gain (Järvelin & Kekäläinen, 2002) asks whether the steps that *genuinely* matter appear at the top of the ranking. It needs per-time-step ground truth, which only a synthetic generator provides, but it is normalised to $[0,1]$ and is therefore the metric we lean on whenever the two disagree.

3 The Design Space We Sweep

A model in this family is a pair *encoder* + *pooling head*. The encoder consumes $(B,1,T)$ and returns (B,d,T); the head returns the prediction (B,C) and the interpretation (B,C,T). Every component below respects that contract, so any encoder can be paired with any head from a configuration file alone. That is what makes the sweep of Section 4 controlled rather than a collection of separate models.

Table 1: The seven pooling heads as two independent slots. g_j^k is the evidence of step j for class k. "Reduces to baseline" names the parameter setting at which the head becomes conjunctive pooling *exactly*; this is what makes the κ sweep of Section 5 a controlled experiment, and what Section 8 shows is not sufficient in practice. L1–L3 are the limitations of Section 2.2.

Head	**Slot 1: attention**	**Slot 2: reduction over time**	**Reduces at**	**Targets**
Conjunctive (baseline)	shared gate	mean	—	—
Class-wise conjunctive	per-class gate	mean	$a_j^k = a_j$	L1
Softmax conjunctive	per-class score	softmax, learnable τ	$\tau \to \infty$	L2
Adaptive class-wise	per-class gate	soft-argmax, learnable β_k	$\beta_k \to 0$	L3
Top-k conjunctive	per-class gate	mean of the k largest	$\kappa = 1$	L3
Attention-max	per-class gate	blend of mean and max	$\lambda_k \to 0$	L3
Per-class gated attention	gated $\mathtt{tanh}\odot\sigma$	softmax	**none**	L1, L3
Dual-stream conjunctive	gate + critical query	blend of two streams	$\lambda \to 0$	L3

3.1 The encoder axis: capacity at fixed structure

We use SEA-Net, a length-preserving convolutional encoder: a stem convolution followed by *multi-scale separable residual blocks*. One unit sums three depthwise convolutions of kernel size $\{5, 11, 23\}$ at a shared dilation, mixes channels with a pointwise 1×1 convolution, and is wrapped in BatchNorm, ReLU, dropout and a residual connection. Dilation grows geometrically and is capped at 16. Three properties matter for this study, and all three are constraints rather than features. Padding is "same" with no stride and no pooling, so T is preserved at every stage – required, because the interpretation has one value per step and AOPCR masks individual steps. The dilation cap keeps each embedding tied to a bounded window, so importance maps stay local. And splitting each convolution into a depthwise and a pointwise step (Chollet, 2017; Howard et al., 2017) costs $dk + d^2$ weights instead of d^2k, which is what makes a small setting reachable at all.

The encoder is instantiated at two capacities, and Section 6 turns that into the paper's capacity control: **narrow** ($d = 64$, 4 blocks, 41–70 K parameters) and **wide** ($d = 128$, 6 blocks, ≈ 269 K parameters). Seven variants sit on top of the same block – bottleneck, gated, input-gated, spike/trend, reconstruction-residual, and two wrappers (multi-scale input channels, multi-scale pyramid). They are described in Section A; for this paper they matter only as the spread of the encoder axis, which Section 6 compares against the spread of the head axis.

3.2 The aggregation axis: seven heads in two slots

Every head keeps the two-branch template of Equation (1) – an attention branch deciding *where* to look and a per-step classifier deciding *what* each position argues for – so a head is fully specified by two choices: the shape of the attention branch (slot 1) and the operator that reduces the T evidence values g_j^k to one number (slot 2). Table 1 lays the seven out on those two axes.

Two of the eight rows are adapted from histopathology MIL rather than invented here: gated attention is due to Ilse et al. (2018), and the dual-stream mechanism to Li et al. (2021). Our contribution to those two is the transfer to time series, the per-class formulation, and – for the dual stream – removing its contrastive pre-training and adding the learnable blend that gives it the reduction property in the fourth column.

Top-k conjunctive pooling, in full. Section 5 rests on this head, so we give it here. Let $g_j^k = a_j^k \mathbf{y}_j^k$ with a_j^k the per-class gate. Instead of averaging all T evidence values, average only the k largest, with k a fixed fraction κ of the series length:

$$k = \max\big(1, \lceil \kappa T \rceil\big), \qquad \hat{y}^k = \frac{1}{k} \sum_{j \in \mathcal{S}_{\text{top}}^k} g_j^k, \qquad \mathbf{I}_{k,j} = g_j^k, \tag{4}$$

where $\mathcal{S}_{\text{top}}^k$ holds the indices of the k largest g_j^k. Note that κ is a hyperparameter, not a weight: this head has *exactly the same parameter count* as class-wise conjunctive pooling.

Property 1 (Exact reduction)**.** *At $\kappa = 1$, Equation (4) is class-wise conjunctive pooling, term for term. For $\kappa < 1$ the gradient reaches only the selected steps.*

Property 1 is the reason Section 5 can attribute its result to selection and to nothing else: the $\kappa = 1$ row of that experiment is not a different model, it is the same model with the selection switched off.

3.3 TRAINING OBJECTIVE

Every model is trained with cross-entropy on the bag logits plus one term that exists for the explanation rather than for the classification:

$$\mathcal{L} = \mathrm{CE}(\hat{y}, y) \;+\; \lambda_{\text{focus}}\Big(-\textstyle\sum_j \bar{a}_j \log(\bar{a}_j + \epsilon)\Big), \tag{5}$$

with $\bar{a}_j$ the class-averaged attention. Attention spread evenly over the series has high entropy, so adding entropy to a minimised loss pushes the model towards concentrated attention. Without it, a model can be right while its importance map is flat, and a reader cannot tell where the evidence stops. This is the second, independent selection knob used in Section 5. We set $\lambda_{\text{focus}} = 0.01$ for every SEA-Net model and 0 for the MILLET baseline, since the term is our addition; label smoothing is 0.13 throughout.

4 EXPERIMENTAL PROTOCOL

Datasets. WebTraffic (Early et al., 2024) is synthetic – $1{,}000$ series of length $1{,}008$ over 10 classes – and its generator records which time steps it modified to create each class. It is therefore the only dataset here on which NDCG@n can be computed at all, and every claim about explanation *correctness* in this paper rests on it. We also use the 85 UCR datasets (Dau et al., 2019) for which MILLET published per-dataset results, and run the remaining 43 so that no conclusion depends on a convenient subset.

One configuration for everything. The central methodological choice of this study is that the baselines are *re-trained here* rather than quoted. A published accuracy mixes the effect of the architecture with the effect of the training recipe its authors used, and the two cannot be separated afterwards. Under one identical configuration they can. Every model in this paper – ours, MILLET, FCN and ResNet (Wang et al., 2017) – is trained with Adam (Kingma & Ba, 2015) at learning rate 1.25×10^{-3}, weight decay 1.2×10^{-4}, at most 400 epochs with patience 60, batch size $\min(\lfloor n_{\text{train}}/10\rfloor, 16)$, and a stratified $80/20$ validation split whenever a dataset has at least 100 training series. The full list is in Section B. The price of this choice is that our numbers are not directly comparable with MILLET's published ones; that comparison is a separate question, answered on its own in Section F, and Section 7 shows why keeping the two apart is not merely tidiness.

Two things to hold against every result that follows. First, **WebTraffic was the screening set**. Each new encoder and head was evaluated there before the expensive UCR runs, and only the strongest went forward. The WebTraffic numbers in Section 6 therefore carry a selection effect, and the UCR table in the same section is the honest counterweight rather than an afterthought. Second, **seed coverage is uneven**: four models were trained with three seeds and the other 68 with one. Where three seeds exist the spread is not negligible – accuracy standard deviation ranges from 0.003 to 0.030, and AOPCR is proportionally noisier still (± 0.884 for the baseline). We therefore adopt one rule and apply it in both directions throughout: *a difference smaller than the relevant seed spread is reported as noise, not as a result*, and every single-seed number is marked as such.

Reporting. Classification is measured by accuracy, explanation quality by AOPCR and NDCG@n, cost by parameter count, model size, FLOPs and time per prediction. Over the UCR archive we report the mean accuracy *and* the mean accuracy rank across datasets, following Demšar (2006), because a mean over datasets this heterogeneous can be dominated by a few of them while a rank cannot; ranks are computed over the 84 datasets shared by all fully-swept models.

5 SELECTION INSIDE THE AGGREGATOR DRIVES EXPLANATION QUALITY

Finding 1. *Restricting the bag score to a subset of per-step evidence improves the ground-truth ranking quality of the explanation, at a cost of roughly one accuracy point. The effect is caused by the selection itself, not by the encoder, and it has an optimum rather than growing without limit.*

Table 2: Top-k fraction κ, on WebTraffic. The model is the SEA-Net bottleneck encoder with Top-k pooling and only κ changes; $\kappa = 1.00$ recovers class-wise conjunctive pooling exactly (Property 1). **All five rows are a single seed** (seed 0), so the 0.938/0.940 step is inside the seed spread of this model (± 0.016) and is not read as a difference.

κ	Accuracy	Test loss	AOPCR	NDCG@n
0.05	0.888	0.401	2.288	0.715
0.10 *(default)*	0.938	0.299	2.778	**0.777**
0.25	**0.940**	**0.277**	2.648	0.733
0.50	0.882	0.418	**3.114**	0.732
1.00 *(= class-wise conjunctive)*	0.908	0.395	2.436	0.722

Table 3: The attention-entropy term of Equation (5), on and off, on WebTraffic. Same encoder, same head, same κ, same seed (0); only λ_{focus} changes. One matched pair, so this indicates a direction rather than a measured effect size.

λ_{focus}	Accuracy	Test loss	AOPCR	NDCG@n
0.01 *(on, our default)*	0.938	0.299	**2.778**	**0.777**
0.00 *(off)*	**0.950**	**0.263**	2.303	0.765

5.1 A CONTROLLED EXPERIMENT ON SELECTION

Most comparisons between pooling heads confound the aggregation rule with everything else that changed. Top-k conjunctive pooling avoids that, because of Property 1: at $\kappa = 1$ it *is* class-wise conjunctive pooling, term for term, and κ adds no parameters. Sweeping κ therefore holds the encoder, the head, the parameter count, the objective and the seed fixed, and varies one thing – how much of the evidence the prediction is allowed to use.

Table 2 shows three things, and we state the third because it complicates the first two.

Selection helps, and the mechanism is identifiable. NDCG@n is 0.777 at $\kappa = 0.10$ and falls to 0.722 at $\kappa = 1.00$, an improvement of 0.055 obtained with zero extra parameters. The mechanism follows directly from Equation (4): averaging only the k largest evidence values leaves $\mathbf{I}_{k,j}$ exactly zero wherever the model used nothing, instead of a small weight that never quite reaches zero. NDCG@n scores whether the genuinely modified steps sit at the top of the ranking, so removing the background from that ranking is precisely what the metric rewards. Accuracy at $\kappa = 1.00$ is 0.908, clearly below the 0.938–0.940 obtained at $\kappa = 0.10$–0.25, so the advantage does not come from the encoder either.

There is an optimum, not a monotone gain. At $\kappa = 0.05$ both quantities collapse – accuracy to 0.888 and NDCG@n to 0.715, *below* the $\kappa = 1$ baseline. Selecting too few steps discards evidence the classifier needs, and the explanation degrades with it. The useful statement is therefore not "sparser is better" but "there is a band, and on this dataset it is near $\kappa = 0.1$".

AOPCR does not track κ. It runs $2.288 \rightarrow 2.778 \rightarrow 2.648 \rightarrow 3.114 \rightarrow 2.436$, peaking where NDCG@$n$ is mediocre and the accuracy is worst. Two readings are available: either faithfulness-to-the-model and correctness-against-ground-truth genuinely diverge here, or AOPCR is too poorly conditioned to resolve a change this size. Section 7 gives direct evidence for the second, which is why Finding 1 is stated in terms of NDCG@n.

5.2 A SECOND, INDEPENDENT KNOB POINTS THE SAME WAY

If selection is really the operative variable, then a different mechanism that also concentrates the evidence should move the same quantities in the same direction. The attention-entropy term of Equation (5) is that mechanism: it acts on the objective rather than on the aggregation rule, and it is switched by one hyperparameter.

Switching the penalty off improves classification – 0.950 against 0.938, at a lower test loss – and degrades both explanation metrics. Two different interventions, one in the aggregation rule and

Table 4: WebTraffic, ranked by accuracy, with the re-trained baselines below. S is the number of seeds; $S = 3$ rows give the mean and the per-seed values are in Section D, $S = 1$ rows are a single run. Rows marked † pair *our* encoder with *MILLET's* head and are included so the two halves can be separated. AOPCR is unnormalised (Section 7): this column may be compared down its own rows and nowhere else. The full 72-model table is in Section G.

Model (encoder + head)	Accuracy	AOPCR	NDCG@n	Params	S
SEA-Net gated (narrow) + Top-k	**0.955**±0.003	2.225	0.750	61,740	3
SEA-Net wide + conjunctive†	0.954	1.502	0.698	269,083	1
SEA-Net bottleneck (narrow) + Top-k	0.947±0.016	2.621	**0.773**	**41,324**	3
SEA-Net wide + class-wise conjunctive	0.946	1.722	0.686	269,164	1
SEA-Net input-gated (narrow) + adaptive	0.905±0.030	**2.651**	0.732	58,102	3
Baselines, re-trained here under the identical configuration of Section 4					
MILLET (InceptionTime + conjunctive)	0.887±0.010	2.569	0.677	423,707	3
ResNet + conjunctive	0.772	2.952	0.554	506,331	1
FCN + conjunctive	0.742	3.826	0.533	267,035	1

one in the loss, therefore produce the same trade: about one accuracy point in exchange for a more concentrated and better-ranked explanation. That consistency is the evidence that the trade is a property of concentrating evidence, rather than an artefact of either mechanism.

The practical reading. A designer of an inherently interpretable classifier has a dial here, and it sits in the aggregator and the objective, not in the backbone. Turning it costs about a point of accuracy and buys an explanation whose support is explicit. Whether that is worth paying is an application question, but it is now a priced one. The obvious caveat is that both experiments are single-seed on one dataset; with a per-model spread of ± 0.016 we read the large steps and refuse the small ones, and we did not have the compute to repeat them.

6 Faithful Interpretability Is Cheap, and the Remaining Gap Is Capacity

Finding 2. *A* 41 *K-parameter model matches or exceeds a* 424 *K-parameter MILLET baseline on accuracy, AOPCR and NDCG@*n *at the same time. Where the small models do fall behind – the UCR archive – a wider version of the same encoder overtakes the baseline, which places the effect in capacity rather than in architecture.*

6.1 Where explanation correctness can be checked

WebTraffic is the only dataset here with per-time-step ground truth, so it is the only place all three quantities can be read at once.

The row that carries Finding 2 is the third: at 41 K parameters it is the smallest model in the table and simultaneously holds its best NDCG@n (0.773 against the baseline's 0.677), a higher accuracy (0.947 against 0.887) and a comparable AOPCR (2.621 against 2.569, well inside the baseline's own ± 0.884 spread). Nothing was traded: the reduction did not cost classification quality *or* explanation quality. We deliberately do not call the 0.955 model the winner – the gap to 0.947 is half of the second model's seed spread, and by the rule of Section 4 that is not a result.

The comparison with the two wide rows is more informative than the ranking. Ranks two and four spend about 269 K parameters, six times the third row's budget, to land within 0.007 accuracy of it – and both score *worse* on both explanation metrics. On this dataset, capacity bought neither accuracy nor faithfulness.

6.2 Cost, including the parts that do not favour us

Against the baseline, the 41 K model uses $10.3\times$ fewer parameters, $11.1\times$ fewer FLOPs and a $2.9\times$ smaller saved file (1.43 MB against 4.11 MB), measured on the real $1{,}008$-step input. Three

Table 5: UCR: mean accuracy and mean accuracy rank, all re-trained here under the identical configuration except the first row. Lower rank is better; ranks are over the 84 datasets shared by all 28 fully-swept models. † = our encoder with MILLET's head. The last row is the same architecture as the baseline under *MILLET's own* 1500-epoch recipe, and is included to size the effect of the training budget against the effect of architecture.

Model	**Params**	**Mean accuracy**	**Mean rank**
Wide encoder (≈ 269 K)			
SEA-Net wide + MILLET additive†	269,083	**0.8298**	11.98
SEA-Net wide + class-wise conjunctive	269,164	0.8292	**11.53**
SEA-Net wide + MILLET conjunctive†	269,083	0.8238	12.43
Narrow encoder (41–62 K), the models of Table 4			
SEA-Net input-gated + adaptive	58,102	0.8153	15.47
SEA-Net bottleneck + Top-k	41,324	0.8097	15.40
SEA-Net gated + Top-k	61,740	0.8083	15.29
Baselines			
MILLET (InceptionTime + conjunctive)	423,707	0.8274	12.60
ResNet + conjunctive	506,331	0.8146	13.54
FCN + conjunctive	267,035	0.8141	14.48
MILLET, *their* 1500-epoch recipe, our code	423,707	0.8434	**9.41**

qualifications belong in the same sentence, not in an appendix. Prediction latency improves by only $1.55\times$ (0.131 ms against 0.203 ms), because depthwise-separable convolutions trade parameters for memory traffic rather than for time. Peak activation memory is *worse* (179 MB against 112 MB at batch 32): fewer weights did not mean a smaller footprint. And the SEA-Net models take two to three times longer to reach convergence (117 ± 14 s against 50 ± 10 s). The claim supported here is a reduction in parameters, FLOPs and stored size – which is the budget that matters for shipping a model to a constrained device – and not a general claim of efficiency. Full measurements are in Section C.

6.3 The UCR archive: where the small models lose, and why

One dataset cannot support a general claim, so the same models were run over the 85 UCR datasets for which MILLET published results.

Table 5 separates two effects that Table 4 cannot.

The architecture is not the problem. At comparable capacity, the SEA-Net encoder with our class-wise head reaches mean rank 11.53 against the baseline's 12.60, with 36% fewer parameters, and the two † rows show the encoder also carries MILLET's own heads to 11.98 and 12.43. Whatever costs the narrow models their 15.3–15.5 ranks, it is not the block design.

Shrinking is the problem, and we can price it. Going from 269 K to 41–62 K parameters costs three to four rank places and about 0.02 mean accuracy. On WebTraffic the same reduction cost nothing at all (Section 6.1). A single synthetic dataset with a clear generative structure simply does not need the capacity that 85 heterogeneous real datasets do, and a study that stopped at WebTraffic – as the screening protocol of Section 4 would have allowed – would have reported a general result that is in fact dataset-specific.

The training budget outweighs everything we changed. The last row is the *same architecture and the same code* as the baseline, differing only in the recipe, and it moves the mean rank by 3.19 places ($12.60 \rightarrow 9.41$). The best architectural change in this entire study moves it by 1.07 ($12.60 \rightarrow 11.53$). We could not re-run our own encoders under the longer schedule, so we cannot say how much of that 3.19 would transfer; what the comparison does establish is that conclusions drawn from architecture comparisons at a fixed short budget are conclusions about that budget as much as about the architectures. It also confirms that our re-implementation is faithful: under MILLET's own hyperparameters it reproduces their published UCR mean to within a thousandth (0.8434 against 0.8445, Section F).

Table 6: The same architecture and the same code under two training recipes. Accuracy moves by a few points, as expected from a longer schedule. AOPCR moves by a factor of five, on both datasets.

	WebTraffic		UCR-85	
Training recipe	**Accuracy**	**AOPCR**	**Accuracy**	**AOPCR**
Ours, 400 epochs (Section 4)	0.887	2.569	0.8274	0.8119
MILLET's, 1500 epochs	0.920	**13.268**	0.8434	**3.9264**
ratio	$1.04\times$	$\mathbf{5.17}\times$	$1.02\times$	$\mathbf{4.84}\times$

6.4 Neither axis dominates the other

Finally, the sweep answers the question that motivated it. Averaged over every partner it was paired with, the encoder choice moves WebTraffic accuracy over 0.771–0.937 and the pooling-head choice over 0.744–0.921 (Section E) – spreads of 0.166 and 0.177, effectively the same size. The head is not a detail attached to a backbone; it is an axis of comparable weight. Moreover, the best pair is not built from the two individually best halves: the gated encoder averages 0.902 and Top-k pooling 0.904, both mid-table, yet their combination is the most accurate model we trained. Encoder and head interact, so the combination has to be searched rather than composed – which is the practical argument for building the sweep in the first place.

7 AOPCR Is Not Comparable Across Training Configurations

Finding 3. *Holding the architecture, the code, the metric implementation and the data fixed, and changing only the training recipe, moves AOPCR by a factor of five. AOPCR therefore cannot support comparisons between models trained under different configurations – which is what a comparison across papers is.*

7.1 The measurement

The comparison in Table 6 is unusually clean, because both rows are literally the same model. Same InceptionTime encoder, same conjunctive head, same parameter count, same data loaders, same AOPCR implementation, evaluated in the same harness on the same machine. The only difference is the training recipe: our 400-epoch configuration of Section 4 against MILLET's own published 1500-epoch recipe.

A $5\times$ change in a metric that is supposed to measure explanation quality, produced by a change that improved accuracy by three points, is not a $5\times$ improvement in explanation quality.

7.2 Why this happens

The cause is visible in Equation (3). AOPC accumulates *raw confidence differences* $\hat{y}^k(\mathbf{X}) - \hat{y}^k(\mathbf{X}_j^{O^k})$, and AOPCR takes the margin of that quantity over random maskings. Neither operation normalises by the range the confidence can move over. So the metric inherits the scale of the model's outputs, and anything that changes that scale changes AOPCR without changing the ranking it is meant to evaluate. A longer schedule with no weight decay and no label smoothing – MILLET's recipe – produces exactly that: larger, more saturated logits, hence larger differences when a step is masked, hence a larger area. The ordering of the time steps, which is the thing an explanation actually asserts, need not have improved at all.

This also explains the behaviour noted in Section 5.1, where AOPCR failed to track κ while NDCG@n did. A metric whose scale is set by the confidence distribution will be a noisy instrument for changes that act on the ranking.

Table 7: WebTraffic accuracy, AOPCR and NDCG@n for each pooling head, averaged over the n encoders it was paired with. "Reduces" repeats the fourth column of Table 1. The κ-ablation configurations and the alternative-recipe baseline are excluded, so that no pairing is counted twice. Groups with small n are not comparable with groups averaged over ten or more pairings; the full grid is in Section E.

Pooling head	**Reduces**	**Mean acc.**	**Mean AOPCR**	**Mean NDCG@n**	n
Class-wise conjunctive	yes	**0.921**	2.322	0.692	10
Per-class gated attention	**no**	0.918	1.869	**0.733**	4
Adaptive class-wise	yes	0.912	2.210	0.732	9
Softmax conjunctive	yes	0.907	1.690	0.712	9
Top-k conjunctive	yes	0.904	2.267	0.712	16
Attention-max	yes	0.838	1.758	0.645	8
Dual-stream conjunctive	**yes**	**0.744**	2.007	0.636	4

7.3 What follows for practice

We are not arguing that AOPCR should be abandoned. It is the only faithfulness measure available on datasets without per-step ground truth, which is nearly all of them, and *within* a fixed configuration it does its job. We argue that three things should accompany it, none of which costs anything:

1. **Report the training configuration next to the AOPCR**, and compare AOPCR only within one. A table of AOPCR values whose rows come from different papers has no defined meaning.
2. **Re-train the baseline rather than quoting it**, when the claim involves a faithfulness metric. This is the reason every baseline in this paper was re-trained, and Table 6 is the evidence that the extra compute was not optional.
3. **Prefer a normalised metric when ground truth exists.** NDCG@n is bounded in $[0, 1]$ and is unaffected by logit scale; where the two disagree it is the stronger evidence.

Applied to ourselves, this is why Table 4 carries a warning on its AOPCR column, and why we never place any AOPCR in this paper beside the value MILLET published. Applied to the literature, it means that published AOPCR comparisons between independently trained models – including comparisons we could have made in our own favour – should be read as uninformative rather than as evidence. A normalised replacement that preserves AOPCR's ground-truth-free property is the natural next step, and we did not have the compute budget to validate one here.

8 Capacity-Safety Is Not Optimisation-Safety

Finding 4. *Five of our seven heads provably reduce to the conjunctive baseline for one setting of their parameters. That property is routinely used to argue a design "cannot hurt". One such head is nevertheless the worst we tested, by a wide margin, and the reason is that no term in the objective drives the model back to the safe branch.*

The fourth column of Table 1 is an argument that appears often in the attention and MIL literature: if a new head contains the established one as a special case, then in the worst case the optimiser recovers the established one, so adding the head is free. Table 7 tests that argument on eight heads averaged over every encoder they were paired with.

The counterexample. Dual-stream conjunctive pooling blends the class-wise conjunctive mean with a critical-instance stream through a single learnable scalar $\lambda = \sigma(\theta)$, initialised at 0.05. At $\lambda \to 0$ it is the baseline exactly. It is nonetheless the weakest head in the study: mean accuracy 0.744 against 0.90 and above for the three strongest heads, with its four pairings at 0.860, 0.824, 0.736 and 0.556. A head that provably cannot be worse than 0.921 scored 0.556 on one pairing.

The mechanism. The property is about *representational capacity*; the failure is one of *optimisation*. Nothing in Equation (5) rewards small λ. If the critical-instance query learns a poor embedding early in training, the gradient through the second stream is not informative, but there is also no pressure returning λ towards zero – the model sits in a region where the blend is harmful and the safe branch, though reachable in principle, is never searched for. The model *can* represent the baseline; it never finds it. This is not specific to the dual stream: any "contains the baseline as a special case" argument has the same gap between what the parameter space allows and what the optimiser visits.

The contrast that supports the reading. Two comparisons inside Table 7 sharpen it. Top-k pooling has the same reduction property but implements it through a *hyperparameter* rather than a learned weight: at $\kappa = 1$ the baseline is recovered by construction, not by optimisation, and Top-k never collapses. Conversely, per-class gated attention is the only head with *no* reduction at all, and its behaviour is the mirror image – a respectable mean of 0.918 and the best mean NDCG@n, but never the best partner for any encoder. It has no known-good configuration to fall back to, so it neither wins nor fails outright.

What a designer should take from this. A reduction property is worth having, but it should be built where the optimiser cannot lose it: prefer a fixed hyperparameter over a learned blend, initialise a learned blend at the safe value so that leaving it is a deliberate move, or add an explicit penalty holding the blend near the safe branch early in training. We initialised λ at 0.05 rather than 0, which we now regard as the error; we did not have the compute to re-run the alternative, so this is a diagnosis with a proposed remedy rather than a demonstrated fix.

9 Related Work

Deep time series classification. Wang et al. (2017) established FCN and ResNet as strong convolutional baselines, and InceptionTime (Ismail Fawaz et al., 2020) applies filters of several kernel sizes in parallel, remaining the reference model of the family (Ismail Fawaz et al., 2019; Dau et al., 2019). Dilated convolutions (Bai et al., 2018) and depthwise-separable factorisation (Chollet, 2017; Howard et al., 2017) are the two ingredients our encoder reuses. All of these end in global average pooling, which collapses the time axis and removes the per-step information an inherent explanation needs – which is the structural reason the MIL route exists.

Post-hoc explanation. LIME (Ribeiro et al., 2016), SHAP (Lundberg & Lee, 2017) and Integrated Gradients (Sundararajan et al., 2017) explain a frozen model afterwards. They add inference cost and return an approximation that can disagree with the computation it describes, which is the argument of Rudin (2019) for interpretable-by-design models and the premise of this line of work.

Interpretable-by-design time series models. One direction learns shapelets: VQShape (Wen et al., 2024) represents a series as tokens from a learned codebook of abstracted shapes, and InterpGN (Wen et al., 2025) gates a shapelet expert against a deep expert. The other, which we study, is MIL: MILLET (Early et al., 2024) is its reference point, built on attention-based MIL (Ilse et al., 2018) from histopathology, where multi-scale fusion (Hashimoto et al., 2020) and dual-stream aggregation (Li et al., 2021) originate. Our contribution to this literature is not another head but a controlled comparison across the whole design space, and the observation that the aggregator carries as much of the outcome as the backbone.

Evaluating explanations. AOPC (Samek et al., 2017) and NDCG@n (Järvelin & Kekäläinen, 2002) are the two measures used here, in the forms adopted by Early et al. (2024). Section 7 adds a measured caveat to the first of them. The concern is adjacent to a wider one about perturbation-based faithfulness measures – that they conflate the explanation with the model's sensitivity to masking – but our finding is narrower and more mechanical: the metric is unnormalised, so the training recipe alone moves it by more than any method we compared.

10 Limitations

Five bounds apply to everything above. **Explanation correctness is measured on one synthetic dataset**, because NDCG@n needs per-step ground truth and only a generator supplies it; how these results hold on real, noisier signals is untested, and constructing a second ground-truth dataset is the single most valuable extension we can name. **Finding 1 rests on single-seed runs**: with a per-model spread of ± 0.016 accuracy we read only the large steps in Table 2 and refuse the small ones, but repeated runs would settle the shape of the curve properly. **The headline models were selected on the dataset they are headlined on** (Section 4), which is exactly what Table 5 exists to bound. **Only univariate series were tested**; the encoder is indifferent to input channel count, so multivariate input should work mechanically, but that was not verified. **Faithfulness is not usefulness**: AOPCR and NDCG@n both ask whether the stated explanation matches something checkable, and neither establishes that a person would find the highlighted steps informative – that requires a human study.

11 Conclusion

We treated a family of inherently interpretable time series classifiers as a design space rather than as a sequence of models, and swept it under one configuration. Three of the four results point the same way: the aggregator matters more than its reputation suggests. Selection inside the aggregation rule, not encoder capacity, is what makes an explanation rank the right time steps, and it costs about one accuracy point (Section 5). The head axis moves accuracy as much as the encoder axis, and the two interact, so the pair has to be searched (Section 6). And a head's reduction property protects it only if the optimiser is given a reason to use it (Section 8).

The fourth result is a caution about how this work is measured. AOPCR moves by $5\times$ under a change of training recipe alone (Section 7), which is larger than any difference between the methods we compared. Until a normalised replacement exists, faithfulness claims in this literature should be made within a single training configuration, against baselines re-trained inside it. That is a heavier experimental burden than quoting a published number, and Table 6 is our argument that it is not optional.

Finally, interpretability turned out to be cheap and capacity turned out not to be: a 41 K model matched a 424 K baseline on accuracy and both explanation metrics simultaneously, while the same architecture widened back up recovered the archive-wide performance the small version lost. The gap that remains against the best result we measured is not an architectural one at all – it is a training budget of 1500 epochs against 400.

Reproducibility Statement

Every number in this paper was produced by one pipeline and can be regenerated from it. The training configuration shared by all 72 model variants is summarised in Section 4 and given in full, including the head-specific initial values, in Section B. The datasets are public: the UCR archive (Dau et al., 2019) and the WebTraffic generator released with Early et al. (2024). Each model is a single configuration file rather than a script, so every row of every table corresponds to one named configuration; the mapping from table row to configuration identifier is given in Section G. Per-dataset results, per-seed results, the cost measurements of Section C and the code that produced every figure and table are in the anonymised repository at https://anonymous.4open.science/r/seanet-ANON. Two limits are worth stating plainly here rather than leaving to be discovered: 68 of the 72 variants were trained with a single seed (Section 4), and the κ and entropy ablations of Section 5 are single-seed matched pairs.

Ethics Statement

This work uses only public benchmark datasets and one synthetic generator; no human subjects, no personal data, and no new data collection are involved. The paper's subject matter carries one risk worth naming: it concerns explanations, and an explanation that is faithful to a model is not thereby correct about the world. Section 7 and Section 10 argue that the metrics in current use are weaker evidence than they appear, and we would not want a result from this paper cited to support deploying an interpretable classifier in a high-stakes setting without a human evaluation of the explanations it produces. The code we build on is released under the Apache License 2.0 and its attribution notice is preserved in our release.

Use of AI Tools

An AI coding assistant was used during this project for three things: writing and refactoring parts of the experiment-management code (configuration handling, results aggregation, figure and table generation), as an aid while drafting and editing the text of this paper, and for routine proof-reading. It was not used to design the models, choose the experiments, run any training, or generate, select or interpret any result. Every number reported here was produced by executing the released code on the datasets named in Section 4, and every claim, table and figure was checked against the stored result files by the authors, who take full responsibility for the contents of this paper.

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A THE SEA-NET ENCODER VARIANTS

All seven variants share the multi-scale separable block of Section 3.1 and change it at one point. Each is stated as the problem it addresses and the modification.

Bottleneck. The pointwise 1×1 step dominates the parameter count, costing d^2 per unit. Factorise it through a narrow middle layer of width $r = d/4$: $P = P_{\text{expand}} P_{\text{squeeze}}$, cost $d^2 \to 2dr = d^2/2$. At $d = 64$ with 4 blocks this takes the whole model from 57,580 to 41,324 parameters, a 28% saving on the encoder. This is the smallest encoder here and the one in the headline row of Table 4.

Gated. Not every channel is informative for every series. Collapse the time axis to a summary $\mathbf{s}$, form a gate $\mathbf{g} = \sigma(W_g \mathbf{s}) \in [0,1]^d$, and re-weight every step by $(1 + \mathbf{g})$. The multiplier is $(1 + \mathbf{g}) \in [1, 2]$ rather than $\mathbf{g}$ deliberately: a plain product could drive features to zero, whereas amplifying on top of the original preserves the differences between time steps, which is what AOPCR and NDCG@n measure. Three summaries (max, mean, last) were trained.

Input-gated. The gate above is one vector for the whole series. Compute it instead from the raw input with a kernel-7 convolution, giving $\mathbf{g}_j$ per time step. The two variants differ only in the subscript, so training both isolates the granularity of gating.

Spike/trend. Convolution plus normalisation is biased towards smooth patterns. Run an explicitly high-pass branch – subtract a moving average of window 5, convolve to 16 channels – and concatenate it with the trend features. The expectation was an advantage on impulsive evidence; Table 11 shows it was the weakest true encoder, so the expectation was not met.

Reconstruction-residual. Rather than assuming what the encoder misses, measure it: a 1×1 convolution reconstructs the raw input from the trend features, and the residual $\mathbf{X} - \widehat{\mathbf{X}}$ becomes 16 extra channels. There is no separate reconstruction loss, so this is a learned residual feature and not an auto-encoder. It was the second strongest encoder.

Multi-scale channels (wrapper). Append rolling maximum, minimum and standard deviation over windows $\{3, 7, 15, 31\}$ plus the sign of the first difference, widening the input from 1 to 14 channels. Max, min and standard deviation are used rather than a rolling mean because a mean is linear and the stem convolution could learn it unaided.

Multi-scale pyramid (wrapper). Run one shared encoder on the series downsampled by $\{1, 2, 4, 8\}$, stretch the outputs back to length T, and fuse with attention over the *scale* axis, never over time – the per-step signal the metrics depend on is preserved. This is the one-dimensional analogue of multi-magnification whole-slide MIL (Hashimoto et al., 2020). It was the weakest variant built, and it imposes a constraint the pipeline checks automatically: the wrapped encoder's receptive field must not exceed $T/8$, or the coarse scales collapse.

Rejected alternatives. A Transformer encoder was rejected on two grounds: full self-attention costs $\mathcal{O}(T^2)$, prohibitive at 1,000–1,500 steps on a constrained device, and unrestricted attention lets any output position draw on any input position, which diffuses the importance map. Pruning or quantising InceptionTime would reduce size but leaves the global average pooling in place, so the model would still need a post-hoc explainer.

B Full Hyperparameters

Table 8: Every setting shared across the study, plus the two encoder capacities. The head-specific initial values are the constructor defaults and are not overridden by any configuration file.

Group	Parameter	Value
Optimisation	optimiser	Adam (Kingma & Ba, 2015)
	learning rate / weight decay	1.25×10^{-3} / 1.2×10^{-4}
	epochs / patience	max 400 / 60
	batch size	$\min(\lfloor n_{\text{train}}/10 \rfloor, 16)$, floor 2
	validation split	stratified 80/20 if $n_{\text{train}} \geq 100$
	label smoothing	0.13
	λ_{focus} (Equation (5))	0.01; 0 for the MILLET baseline
	seeds	3 for four models, 1 for the other 68
Encoder, narrow	width d / blocks	64 / 4
Encoder, wide	width d / blocks	128 / 6
both	kernels $\mathcal{K}$ / dilation cap / dropout	$\{5, 11, 23\}$ / 16 / 0.2
Pooling	attention width d_a / dropout	8 / 0.2
	Top-k fraction κ	0.1
	initial τ_0 (softmax conjunctive)	1.0
	initial β_0 (adaptive class-wise)	0.3
	initial λ_0 (attention-max / dual-stream)	0.5 / 0.05

C Cost Measurements

Table 9: Measured on the real WebTraffic input (length 1,008, 10 classes, batch 32) on one GPU. "Train" is wall-clock time to convergence including early stopping, mean $\pm$ sd over the three WebTraffic seeds where available. The last two columns are the ones that do not favour the small models and are discussed in Section 6.2.

Model	Params	Size (MB)	FLOPs (M)	1 pred. (ms)	Peak mem (MB)	Train (s)
SEA-Net bottleneck + Top-k	**41,324**	**1.43**	**76.7**	0.131	179.0	117.3 ± 14.0
SEA-Net gated + Top-k	61,740	1.50	109.7	0.128	**64.9**	91.3 ± 25.9
SEA-Net wide + class-wise	269,164	3.54	523.7	0.359	123.7	—
MILLET	423,707	4.11	847.6	0.203	112.3	50.4 ± 9.5
ResNet + conjunctive	506,331	4.42	1013.2	0.139	76.5	38.1
FCN + conjunctive	267,035	3.47	535.2	**0.068**	132.2	20.8

D Per-Seed Results

Table 10: WebTraffic accuracy for the four models trained with three seeds. These four spreads are what the rule of Section 4 is calibrated against.

Model	Seed 0	Seed 1	Seed 2	Mean $\pm$ sd
SEA-Net gated + Top-k	0.954	0.958	0.952	0.955 ± 0.003
SEA-Net bottleneck + Top-k	0.938	0.938	0.966	0.947 ± 0.016
SEA-Net input-gated + adaptive	0.938	0.880	0.898	0.905 ± 0.030
MILLET, re-trained here	0.894	0.876	0.892	0.887 ± 0.010

The three behave differently, and this matters for how far each result travels. The gated model is stable, six thousandths between best and worst seed. The bottleneck model is tight on two seeds and jumps on the third, so its mean is raised by one good run. The input-gated model is genuinely dispersed, which is why its AOPCR advantage in Table 4 is read as noise.

E Encoder Ablation

Table 11: WebTraffic accuracy, AOPCR and NDCG@n for each encoder, averaged over the n pooling heads it was paired with; † marks a wrapper. The companion table on the head axis is Table 7. Two cautions: several groups are small, and the bottleneck encoder's mean is dragged down because it was also paired with the weak heads – its best pairing is the headline row of Table 4.

Encoder	Mean acc.	Mean AOPCR	Mean NDCG@n	n
SEA-Net multi-scale channels†	**0.937**	2.385	**0.747**	4
SEA-Net reconstruction-residual	0.924	2.315	0.719	5
SEA-Net input-gated	0.904	2.345	0.718	5
SEA-Net gated	0.902	2.056	0.710	19
SEA-Net base	0.898	1.933	0.694	12
SEA-Net bottleneck	0.884	2.209	0.694	8
SEA-Net spike/trend	0.858	1.814	0.677	7
SEA-Net multi-scale pyramid†	0.771	1.394	0.604	3
InceptionTime (MILLET)	0.887	2.569	0.677	1
ResNet	0.772	2.952	0.554	1
FCN	0.742	3.826	0.533	1

The encoder spread is 0.771–0.937 (0.166) against the head spread of 0.744–0.921 (0.177) in Table 7. Those two numbers are the evidence for Section 6.4.

F Relation to the Published MILLET Results

The main text compares only models re-trained inside this harness. This appendix answers the separate question that choice raises: how do those numbers relate to the ones MILLET published, and is the difference caused by our re-implementation or by the training configuration?

Two conclusions follow. First, the re-implementation is correct: under MILLET's own hyperparameters it reproduces their UCR mean to within about one thousandth (0.8434 against 0.8445), with a balanced per-dataset record of 26 wins, 32 ties and 26 losses against the published numbers, and their released weights evaluated in our harness land inside the spread of their own five seeds. Second, the remaining difference is therefore the training budget, and not the architecture or the code – about 0.016 on the UCR mean and 0.033 on WebTraffic. That figure cannot simply be added to our models, since a longer schedule need not help every architecture equally, but it is the right order of magnitude to hold in mind while reading Table 5. The AOPCR values of these same two runs are what Section 7 is built on.

Table 12: The MILLET baseline (InceptionTime + conjunctive pooling) measured five ways. Only the last row is used anywhere in the main text, except in Table 6 and Table 5 where the fourth row appears explicitly as the alternative recipe.

How the number was obtained	**WebTraffic acc.**	**UCR-85 mean acc.**
Published (Early et al., 2024), mean of 5 seeds	0.924	0.8445
Published (Early et al., 2024), 5-model ensemble	0.940	—
Their released weights, evaluated in this harness	0.922	—
Re-trained here under *their* hyperparameters	0.920	0.8434
Re-trained here under Section 4	**0.887**	**0.8274**

G FULL LEADERBOARD AND PER-DATASET RESULTS

The complete table of all 72 encoder $\times$ head combinations – with the configuration identifier for every row, so that any row can be reproduced by name – together with the full encoder $\times$ head grid, the per-dataset UCR results for every model, the win/tie/loss analysis and the critical-difference diagram, is published with the code at `https://anonymous.4open.science/r/seanet-ANON`. It is omitted here only for length; nothing in the main text depends on a number that appears solely in that material.